

**Fourth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ANALOG AND DIGITAL COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1. (a) An audio frequency signal **$10 \sin (2\pi \times 1000 t)$** used to amplitude modulates a carrier of **$50 \sin (2\pi \times 10^6 t)$** . Determine modulation index, carrier frequency, bandwidth, total power delivered to load of **100Ω** , transmission efficiency.
5(CO1)
 - (b) An equation of frequency modulation wave given by,
 $E_{FM} = 20 \sin (6 \times 10^2 t + 7 \sin 1250 t)$. Determine carrier frequency, modulation frequency, modulation index, power dissipated by FM wave for **100Ω** resistance. Also write an equation of phase modulation.
5(CO1)
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2. (a) Analyze the Noise effect on AM Double Side Band Full Carrier system.
5(CO3)
 - (b) Consider a system with two resistors **$10 K\Omega$** and **$20 K\Omega$** is operated at room temperature of **$17^\circ C$** and bandwidth of **$100 KHz$** . Determine thermal noise voltage for the conditions :
 - (1) For each resistor,
 - (2) For two resistances in series,
 - (3) For two resistances in parallel.5(CO3)

3. (a) The pulse code modulation is used for TV signal transmission with maximum input frequency **4.2** MHz. The number of quantization level **512**. Determine code word length, transmission bandwidth, bit rate and signal to noise ratio in dB. 5(CO1)
- (b) Illustrate the importance of non uniform quantization. Describe in detail an Adaptive delta modulation and its advantages. 5(CO1)
4. (a) Describe MSK Transmitter with mathematical details. 5(CO4)
- (b) Draw waveforms for QPSK Transmitter if input is **0 1 1 0 1 0 0 1**. 5(CO4)
5. (a) Derive an expression for Probability of errors for Amplitude Shift keying. 5(CO2)
- (b) Describe the concept of Inter symbol interference along with Nyquist criterion. 5(CO2)
6. A Convolution Code is described by $X1 = [1 \ 1 \ 1]$; $X2 = [1 \ 0 \ 1]$;
Draw the convolution encoder and state diagram. If the received sequence for above encoder is **0 1 0 0 1 0**, decode the sequence using Viterbi decoding algorithm. 10(CO5)

